

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. SE Branch : CSE Semester :III Subject Code & Name: Data Structure (BTCOC303) Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following. (This is just a sample instruction)		
A)	Define array and show the representation of one-dimensional array is memory.		06
B)	Explain Abstract data types.		06
C)	Write a c code to accomplish the following Stack Operations: i] insert ii]delete iii]display iv]peek		06
Q.2	Solve Any Two of the following. (This is just a sample instruction)		
A)	Write an algorithm to convert infix expression to postfix expression using stack.		06
B)	Explain array and linked list representation of Binary tree.		06
C)	What is queue and explain types of queue.		06
Q. 3	Solve Any One of the following. (This is just a sample instruction)		
A)	Explain representation of graph : A] adjacency matrix representation B] adjacency list representation		06
B)	Explain different file organization in detail.		06
C)	Define and explain the following terms with an example: 1]Path and Cycle 2]Connected , Unconnected graph 3]Complete graph and weighted graph		06
Q.4	Solve Any Two of the following. (This is just a sample instruction)		
A)	What is Depth First Traversal ? Write an algorithm for Depth First Traversals.		06

B)	Write an algorithm for binary search tree . Construct binary search tree for the following list of numbers: 18,17,3,5,9,21,26,15,61,6,8,11.		06
C)	What is Hashing and explain construction of a hash table.		06
Q. 5	Solve Any One of the following. <i>(This is just a sample instruction)</i>		
A)	Explain different file operations.		06
B)	Explain Indexed sequential file in detail.		06
C)	What is graph ? Explain matrix and linked list representation of a graph.		06
	*** End ***		

The grid and the borders of the table will be hidden before final printing.